

*Example 7:* If CRISPR can help parents conceive a disease-free child when no other options exist and it can do so safely, ought we to use it? It's a question I've asked myself again and again [...]

As National Institutes of Health director Francis Collins put it, "Evolution has been working toward optimizing the human genome for 3.85 billion years. Do we really think that some small group of human genome tinkerers could do better without all sorts of unintended consequences?" [...]

While I share the general feeling of unease at the idea of humans taking control of their evolution, I wouldn't go so far as to say that nature has fine-tuned our genetic composition. Obviously, evolution didn't optimize the human genome for the present era, when modern foods, computers and high-speed transportation have completely transformed the way we live. And if we look over our shoulders at the course of evolution that has led to this moment, we'll see it's littered with organisms that didn't benefit from the mutational chaos that underpins evolution. Nature is less an engineer than a tinkerer—and a fairly sloppy one at that. Its carelessness can seem like outright cruelty for those people who have inherited genetic mutations that turned out to be suboptimal. (Doudna and Sternberg 2017)

